BTI/B Tech/MBA Tech
Minor Project/Capstone Project

GUIDELINES

AND

REPORT FORMATS

Approval signature:

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Mukesh Patel School of Technology Management and Engineering
SVKMs NMIMS University

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1 Introduction to the Course

1.1 Preamble

The capstone project aims to provide practical design experience to the students. It will also help the students to acquire the required skill sets in their professional careers. The project guidelines are framed to apprise the students about the procedures to be followed and the expectations to be met. The project guidelines will help supervisors and examiners in evaluation of the project work.

1.2 Course objective

The capstone project is designed to consolidate the learning of the final year students with a hands-on experience. ABET defines a capstone design experience as "a culminating major engineering design experience that 1) incorporates appropriate engineering standards and multiple constraints, and 2) is based on the knowledge and skills acquired in earlier course work. The capstone design course is used to assess the ability to identify, formulate, and solve complex engineering problems, and apply engineering design using ethical and professional responsibilities in engineering situations and working effectively in a team.

1.3 Pre-requisite

- Knowledge of all core and elective courses completed till 3rd year.

2 Course Outcomes (CO) and mapping with Program Outcomes (PO)

2.1 Course Outcomes

After successful completion of the course, a student will be able to -

1. Select an appropriate problem statement after reviewing the literature and identifying the research gaps.
2. Formulate a feasible design model.
3. Implement the prototype/proof of concept, test and validate the results.
4. Work efficiently in a team environment.
5. Summarize the findings into a technical report.

2.2 Program Outcomes

Engineering Graduates will be able to:

1. ***Engineering knowledge***: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. ***Problem analysis***: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. ***Design/development of solutions***: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. ***Conduct investigations of complex problems***: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. ***Modern tool usage***: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

6. ***The engineer and society***: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. ***Environment and sustainability***: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. ***Ethics***: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. ***Individual and team work***: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. ***Communication***: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. ***Project management and finance***: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. ***Life-long learning***: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

2.3 CO-PO mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				1	2	2	2	1	1	
CO2	2	3	3	2	2	1	2	2	2	1	2	1
CO3	2	3	3	3	3	1	2	1	2		2	2
CO4						1	1	1	3	1	3	
CO5									3	1	3	3

3 Teaching and evaluation scheme

Semester VI for BTI programs
Semester VII for all B Tech Programs
Semester VIII for all MBA Tech Programs

Teaching Scheme				**Evaluation Scheme**	
Lecture Hours per week	**Practical Hours per week**	**Tutorial Hours per week**	**Credit**	**Internal Continuous Assessment (ICA) (Marks -50)**	**Term End Examinations (TEE) (Marks - 50)**
0	8	0	4	Marks Scaled to 100	--

4 Guidelines and review process

4.1 Finalization of the project teams and domains

The orientation of the minor project/capstone project, finalizing the project teams and allocation of the mentors is to be completed in the prior semester i.e. semester V/VI/VII as applicable.

Table 4.1.1: Time line for finalization of Capstone project teams and mentors

Week No	Activity	Responsibility and roles

10th week of prior semester (semester V/VI/VII as applicable)	Project Orientation-I	• Department project coordinators.
		  ◦ To introduce the significance, time lines and the expectations of the capstone project as a part of their curriculum.
		  ◦ Each group will be of maximum 2-3 students.
		  ◦ The students should be made aware of the assessment and evaluation policies and the implications of not following the same.

11th week of prior semester (semester VI/VII as applicable)	Submission of project groups and three broad domains	• Students to form project group consisting of 2 to 3 students.
		• The students to submit the following:
		  ◦ Project group details (Interdisciplinary project groups will be permitted based on the problem statement and scope of the project with the approval of the dean)
		  ◦ At least three broad domains in which they intend to work for their capstone project.
		• The submission of the group details and domains to be done as per the template given in annexure.

12th week of prior semester (semester VI/VII as applicable)	Finalization and approval of Project Teams and Mentors	• Department project coordinators in consultation with the head of the department or a committee formed at department level.
13th week of prior semester (semester VI/VII as applicable)	Mentor Allocation to be informed to the students	• Department project coordinators

14th week of prior semester (semester VI/VII as applicable)	Project idea discussion with Mentors	• The project groups to have the first meeting with their respective mentors.
		• Agenda for the meeting:
		  ◦ Briefing to the students about the capstone project and the expectations and requirements by the mentor.
		  ◦ Discussion on the broad domain/ideas submitted by the students and expected outcomes
		  ◦ Action points for the next meeting at the start of the semester:
		    ▪ Literature review on the ideas/problem statement.
		    ▪ Determine research gaps
		    ▪ Prepare a brief synopsis on the finalized topic.

4.2 Project guidance and review time lines

1. The students are expected to meet with the faculty mentor every week (semester VI/VII/VIII as applicable) and update about the work done. Record of the proceeding to be maintained.
2. The faculty mentor is expected to review and evaluate the work done and give suggestions for further work.
3. A google sheet will be maintained by the department project coordinators for recording the status and updates about the weekly interactions of the project groups and mentors and the grading for every week.
4. Student's Project Progress would be reviewed **three times** during the entire semester as per the academic calendar.
5. A separate schedule for submission for various reports is enclosed (Annexure). The student must strictly observe the submission date, while sending reports.
6. For all the project reviews, it is the sole responsibility of the student to report to his faculty mentor, and to make a presentation regarding the project progress in front of the panel of faculty members, which includes the project guide.
7. During all reviews students should represent all technical specification and data of the project. The presentation should be complete in all respect as per the templates given in Annexure to grade it.
8. Any deviation/noncompliance should be reported by the mentor in time to the HoD/Dean.

Table 4.2.1: Minor Project/Capstone project review timelines

Sr. No.	**Review**	**Timeline**
1st review	Topic approval and feasibility	2nd week of semester VI/VII/VIII for BTI/B Tech/MBA Tech
2nd review	Midterm review	7th week of the semester (after midterm test 1)
3rd review	Final presentation and viva	Last week of the semester

5 Assessment Policy

5.1 Component wise Continuous Evaluation Internal Continuous Assessment (ICA)

Assessment Component	**1st review**	**2nd review**	**Progress review**	**3rd review** **Final Presentation and Viva (Internal + External)**
Weightage	10%	25%	35%	30%
Marks	10	25	10 (review-1) + 10 (review-2) + 15 (review-3)	30

5.2 Progress review

All the project team members must meet the faculty mentor regularly and update them about the progress. An individual team member's contribution will be ascertained using different tools.

Continuous assessment of the project work will be carried out based on the fortnightly performance, discussions and reporting to the faculty mentor.

- Log book format will be given to students to maintain a record of work carried out every week and discuss with faculty mentor and get signature of faculty mentor on it. Reporting date and comments will be written on weekly basis in this log book.
- Log book submission to be done by students during the midterm and final presentation.
- Any student not completing the requirements to be reported to the HoD/Dean in time.

The faculty mentors will evaluate the project team for every review on the basis of the rubric shown in table 5.2.1 to 5.2.3.

Table 5.2.1: Rubrics for progress review -1

	Parameter	**Far Exceeds Expectation (5)**	**Exceeds Expectation (4)**	**Met Expectations (3)**	**Below Expectations (2)**
A	Fortnightly project discussion with Mentor (5)	The student reports to the project mentor regularly and consistent in work	The student reports to the project mentor often but not very consistent	The student reports to the project mentor but lacks Consistency	The student is irregular and inconsistent in work
B	Background study (5)	Demonstrates comprehensive understanding of the application domain and problem context.	Good understanding with minor gaps.	Basic understanding with limited depth.	Poor understanding of the domain.
TOTAL MARKS= (A+B)=10					

Table 5.2.2: Rubrics for progress review -2

	Parameter	**Far Exceeds Expectation (5)**	**Exceeds Expectation (4)**	**Met Expectations (3)**	**Below Expectations (2)**
A	Fortnightly project discussion with Mentor (5)	The student reports to the project mentor regularly and consistent in work	The student reports to the project mentor often but not very consistent	The student reports to the project mentor but lacks Consistency	The student is irregular and inconsistent in work
B	Research gaps, Proposed Approach & Technical Justification (5)	Clearly identifies limitations of existing work, defines the research gap, and provides	Identifies the research gap and justifies the proposed	Research gap or technical justification is only partially supported.	No clear research gap or justification for the

		strong justification for the proposed methodology, tools, technologies, and datasets.	approach with minor gaps.		proposed approach.
TOTAL MARKS= (A+B)=10					

Table 5.2.3: Rubrics for progress review -3

	Parameter	**Far Exceeds Expectation**	**Exceeds Expectation**	**Met Expectations**	**Below Expectations**
A	Fortnightly project discussion with Mentor (5)	The student reports to the project mentor regularly and consistent in work (5)	The student reports to the project mentor often but not very consistent (4)	The student reports to the project mentor but lacks Consistency (3)	The student is irregular and inconsistent in work (2)
B	Research Paper Publication (10)	Paper is accepted/published in Scopus indexed Journal/Conference/Book Chapter, with proof of acceptance attached (10)	Paper is submitted with plagiarism less than 10% including AI similarity- with proof of submission attached (8)	Draft paper ready with plagiarism less than 10% including AI similarity (6)	Research Paper Publication (4)
TOTAL MARKS= (A+B)=15					

1st Review Criteria

The student would be evaluated for the first phase of the capstone project as per the time line given in the table 4.2.1. Student has to make the presentation in front of the panel of faculty members that also includes the faculty mentor. It is expected that by this time the project groups have discussed and finalized the topic and the feasibility of the implementation. Following are the guidelines for the 1st Review.

- Each project group to briefly describe the project definition, objectives and the scope of the project.
- The tools and technology which would be used for the design and development of the project to be discussed.
- The students are expected to present the project engagement schedule and the project development time line for the entire semester (Gantt Chart) to be prepared.

- The topic approval will be done based on the rubric shown in table 5.2.4.

Table 5.2.4 Rubrics for 1st review

	Parameter	Far Exceeds Expectation (5)	Exceeds Expectation (4)	Met Expectations (3)	Below Expectations (2)
A	Identification of Problem statement, Purpose and need of the project (5)	Detailed and extensive explanation of the project idea, purpose, need of the project, novelty of the idea and clearly defined objectives	Good explanation of the project idea, purpose and need of the project	Average explanation of the project idea, purpose and need of the project;	No great explanation of the project idea, purpose and need of the project
B	Feasibility of the implementation (5)	Feasibility has been done taking into accounts all the constraints and explained well explained	Has idea about the constraints and considered feasibility	Feasibility has yet to be considered. Has Knowledge about constraints	Not considered constraints and feasibility
	TOTAL MARKS= (A+B)=10				

2nd Review Criteria

The student would be evaluated for the second phase of the capstone project during the 2nd Review as per the time line given in table 4.2.1. Student has to give the presentation in front of the panel of faculty members that also includes the faculty mentor. Following are the guidelines for the 2nd Review.

- Student has to make a presentation covering the Project Progress till date. Student can include the proposed features of the system, the Analysis and design documents as well as the problems related to the system design, tools and technology. Individual contribution to be presented.
- The current progress would be compared with the project engagement schedule and the project development plan, which was submitted during the 1st Review.

After the 2nd review, the Internal Facilitator/Internal Project Reviewer would evaluate the regularity of the student, his knowledge as well as the approach and methodology adopted by him to design and develop the system. His documentation and presentation skills would also be evaluated. If the faculty finds the project progress very slow mentor will be required to to report to the PC/HoD. Student has to demonstrate the working modules of the project in front of the panel of faculty members which includes the faculty mentor.

The second review will be for 20 marks. The components for the evaluation of 2nd review are shown in table 5.2.5.

Table 5.2.5 Rubrics for 2nd review

	Parameter	Far Exceeds Expectation (5)	Exceeds Expectation (4)	Met Expectations (3)	Below Expectations (2)
A	Literature Review/Market Survey (5)	The student has complete understanding of the existing research/products and debates relevant to the project topic or area of study	The student has a good understanding of the existing research/products and debates relevant to the project topic or area of study	The student has conducted a very narrow research on existing systems/methodologies/products.	No remarkable literature/market survey conducted
B	Clarity of the problem statement (5)	The student has identified the problem statement after identifying research gaps and stated the objectives well.	The student stated the objectives well.	The problem statement needs some improvement and the objectives need to be refined.	The problem statement is not clear. Need to be reworked.
C	Design/ System level representation (5)	The system design is very well drawn and gives complete understanding of the proposed model/ Algorithms to be implemented are clearly stated	The system architecture designed gives a good understanding of the proposed model/ Algorithms to be implemented are well stated	The system architecture designed moderately depicts the proposed model/ Algorithms to be implemented are moderately explained	The system architecture is not drawn and there is no complete understanding of the proposed model or Algorithms to be implemented.
D	Implementation/ Results (5)	The student has implemented the model/algorithms and completed more than 50% of the work The results are very well documented and shown in the presentation.	The student has implemented the model/algorithms and completed up to 40% of the work The results are shown in the presentation	The student has started the implementation of the model/algorithms and completed less than 40% of the work Some preliminary results are shown in the presentation	The student has not yet started implementation. Results are not satisfactorily shown in the presentation
E	Interim Report (5)	Complete explanation of the key concepts and strong description of the technical requirements of the project and Future extensions in the project are well specified	Complete explanation of the key concepts but in-sufficient description of the technical requirements of the project Future extensions in the project are specified	Incomplete explanation of the key concepts and in-sufficient description of the technical requirements of the project	Not a great explanation of the key concepts and in-sufficient description of the technical requirements of the project
	Total (A+B+C+D+E) = 25 marks				

3rd review (Final Presentation and Viva)

Student will demonstrate the working modules of the project in front of the external expert and a panel of faculty members formed by the department which includes the faculty mentor. The students shall demonstrate details of the testing, implementation and technical documentation including the outcomes and conclusion. The students should present the overview of all aspects of the project completed including the design/testing/implementation and documentation details.

There will be two components for the final presentation and viva.

1. Internal and External panel member evaluation (30 marks)
   The evaluation for the final presentation and viva will be done based on the rubric shown in table 5.2.6.

Table 5.2.6 Rubrics for internal and external panel member for final Presentation and Viva (3rd review)

Sr. No.	**Parameter**	**Far Exceeds Expectation (5)**	**Exceeds Expectation (4)**	**Met Expectations (3)**	**Below Expectations (2)**
A	Literature review and clarity of the problem statement (5)	The student has identified the problem statement after identifying research gaps and stated the objectives well.	The student stated the objectives well.	The problem statement needs some improvement and the objectives need to be refined.	The problem statement is not clear. Need to be reworked.
B	Design/ System level representation (5)	The system design is very well drawn and gives complete understanding of the proposed model/ Algorithms to be implemented are clearly stated	The system architecture designed gives a good understanding of the proposed model/ Algorithms to be implemented are well stated	The system architecture designed moderately depicts the proposed model/ Algorithms to be implemented are moderately explained	The system architecture is not drawn and there is no complete understanding of the proposed model or Algorithms to be implemented.
C	Implementation and results, attainment of defined objectives (5)	Algorithms/techniques are well stated/modules of project are well integrated and system working is accurate. Defined objectives are attained.	Algorithms/techniques are well stated/ Integration of all modules is not done	Algorithms/techniques are not proper/Modules of project are not properly integrated	No algorithms/ techniques specific to the projects are stated and Modules of project are not integrated
D	Report (5)	Complete explanation of the key concepts and strong description of the technical requirements of the project and Future extensions in the	Complete explanation of the key concepts but in-sufficient description of the technical requirements of the project Future	Incomplete explanation of the key concepts and in-sufficient description of the technical requirements of the project	Not a great explanation of the key concepts and in-sufficient description of the technical requirements of the project

		project are well specified	extensions in the project are specified		
E	Presentation (5)	Contents of presentations are appropriate and well delivered	The project team falters around to answer few questions but The student has thorough knowledge about the project	The project team fails to answer many questions. The student has no great technical insights into the project	The student fail to answer questions. The student has no idea/technical insights into projects
F	Viva (5)	The student is able to answer the questions effectively. The student has thorough knowledge about the project	The student falters around to answer few questions but student has thorough knowledge about the project	The student fails to answer many questions. The student has no great technical insights into the project	The student fails to answer questions. The student has no idea/technical insights into projects
	Total (A+B+C+D+E+F) = 30 marks				

Annexures

A.1 Project registration form

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Project registration form

Department:
Program:
Semester:

Project Team details

S No	SAP ID	Roll No	Name	Signature
1				
2				
3				

Three broad domains in which the project team intends to work

S No	Domain	Project idea in brief
1		
2		
3		

A.2 Topic approval form

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Topic approval form

Department :
Program :
Semester :
Project Team details

S No	SAP ID	Roll No	Name	Signature
1				
2				
3				

Type of Project (Tick any one)

Type of project	
Application	
Product	
Research	

Project details

Project Title	
Project Objectives	
Domain	

Motivation	
Expected outcomes	

Latest References

S No	**Publication (Author, "Title", Journal/Book/Conference, Date, Page, Volume, year)**
1	
2	
3	
4	
5	

Status of the project after Review 1

Comments/Remarks	
Name and Signature of the guide	
Name and signature of panel members for review 1	
Signature of HoD	

If any change in project title after review 1 presentation as per comments received from panel and areas of concern to be noted here

A.3 Evaluation sheet for Review 1: Topic approval and feasibility

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Evaluation sheet for Review 1: Topic approval and feasibility

S No	SAP ID	Roll No	Student Name	Identification of Problem statement, Purpose and need of the project (5 marks)	Feasibility of the implementation (5 marks)	Total (10 marks)

Name & Signature of the Project mentor

(with Date)

Name & Signature of the panel members

(with Date)

A.4 Evaluation sheet for Review 2: Midterm review

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Evaluation sheet for Review 2: Midterm review

S No	**SAP ID**	**Roll No**	**Student Name**	Literature Review/Market Survey **(5 marks)**	Clarity of the problem statement **(5 marks)**	Design/ System level representation **(5 marks)**	Implementation/ Results **(5 marks)**	Interim Report **(5 marks)**	**Total (25 marks)**

Name & Signature of the Project mentor

(with Date)

Name & Signature of the panel members

(with Date)

A.5 Evaluation sheet for Review 3: External Panel Member Evaluation

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Evaluation sheet for Review 3: Internal and External Panel Member Evaluation

S No	**SAP ID**	**Roll No**	**Student Name**	Literature review and clarity of the problem statement **(5 marks)**	Design/ System level representation **(5 marks)**	Implementation and results **(5 marks)**	Report **(5 marks)**	Presentation **(5 marks)**	**Viva** **(5 marks)**	**Total** **(30 marks)**

Name & Signature of the Internal mentor and External Panel Member

(with Date)

A.6 Evaluation sheet for Research paper

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Evaluation sheet for Research paper

S No	SAP ID	Roll No	Student Name	Marks (10 marks)

Name & Signature of the Project mentor

(with Date)

A.7 Log Book for Capstone Project

SVKM'S NMIMS Deemed-to-be UNIVERSITY

Mukesh Patel School of Technology Management & Engineering

Log Book for Capstone Project

Department :
Program :
Semester :
TITLE OF THE PROJECT :

Name of the Faculty Mentor: ______________________________

STUDENT DETAILS

	NAME	ROLL NO.	CONTACT
Student 1			
Student 2			
Student 3			
Student 4			

Name of group members:

Roll No.	Name	Sign(Students)	Grades(By Faculty Mentor)

Week No:
Date of Reporting:
From DD/MM/YYYY to DD/MM/YYYY
Work carried out
Faculty Remark
Faculty Signature with date
Week No:
Date of Reporting:
From DD/MM/YYYY to DD/MM/YYYY
Work carried out
Faculty Remark
Faculty Signature with date

A.8 Project Report Format

Project Report Format

(This page is only for reference and should not be included in the final report)

Follow the given instructions while preparing the report:

1.	Font	Times New Roman.
2	Line spacing	1.5".
3.	Margins (left)	1.5”
4.	Remaining all margins	1”
5.	Header Right corner	Academic Year
6.	Header Left corner	Project Title
7.	Footer: Right Corner	Page Number
8.	Chapter Title	Font size 18 and bold.
9.	Section titles	Font size 14 and bold
10.	Subsection titles	Font size 12 and bold
11.	Figure name (below the figures)	Times new roman , font size 10
12.	Table name (above the table)	Times new roman, font size 10
13.	Report text	Font size 12

14. Appendix will include:

- Specifictions of components
- Pin diagram
- Source code
- Sample codes
- Raw experimental observations etc.
- which shall be numbered in Roman Capitals (e.g. “Appendix IV”).

15. The report (two copies) should be Hard Bound- Black Book

16. The report must be verified and signed by the respective mentors before the final

Presentation / examination.

17.Refer to the following table of contents for the format.

18.Page numbers in Table of Contents for List of Figures, List of Tables and Abbreviations has to be in Roman numbers.

19. Page numbers from Chapter 1. Introduction, onward should be in decimal format.

20. Every chapter should have introduction paragraph and summary lines.

21. Conclusion will be only one, which will be written at the end of report.

Project Title

Project Report submitted in the partial fulfilment

of

(Name of the Program)
In
(Name of the stream)

by

Names (Roll No.)

Under the supervision of

<u>Name of Faculty Mentor</u>

(Designation, Name of the department, MPSTME)

SVKM's NMIMS University

(Deemed-to-be University)

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING (MPSTME)

Vile Parle (W), Mumbai-56

(Academic year)

CERTIFICATE

This is to certify that the project entitled **("Title")**, has been done by **Ms/Mr (name)** under my guidance and supervision & has been submitted in partial fulfilment of the degree of (name of the program) in (name of the stream) of MPSTME, SVKM's NMIMS (Deemed-to-be University), Mumbai, India.

____________	____________
Project mentor (name and Signature)	Examiner (name and Signature)
(Intenal Guide)	

Date

Place: Mumbai

____

(HoD) (name and Signature)

ACKNOWLEDGEMENT

Student Names

NAME	ROLL NO.	SAP ID

ABSTRACT

Approximately one page in which problem shall be defined and outline of proposed work in relation to title of the report shall be given (Times New Roman 12, 1.5 spacing)

What is Abstract?

The abstract for a project is simple, short and can be seen as an overview. It is also like a summary that defines the core of your work. Whether it is literature or science, all papers need some kind of abstract. It will also help the reader understand what you are talking about. Looking at an abstract as a summary will not just make your work easy but also help you write a good one. Always remember that an abstract is not just a summary of the whole paper but also something that could be seen as contributions/conclusions of the work done. To make the abstract more readable thing, make sure to use precise and easily understandable language.

Table of Contents

Conclusion and Future Scope

References

Appendix A: Soft Code Flowcharts

Appendix B: Data Sheets

Appendix C: List of Components

Appendix D: List of Paper Presented and Published.

List of the figures

Fig No	Name of the figure	Page No

List of tables

Table No	Name of the Table	Page No

Abbreviations

Table No		Name of the Table	Page No

Note: Each table should be included on separate page.

Chapter 1

Introduction

Introduction (of the project, and the flow of different chapters)

1.1 Background of the project topic

1.2 Motivation and scope of the report

1.3 Problem statement

1.4 Salient contribution

1.5 Organization of report

Note:

1. **Please include the citations for the papers refered and also include it in the references at the same numbered location as in the text.**
2. **Include the IEEE standards followed for designing the project at appropriate places in the text.**
3. **Minimum number of pages for the report has to compulsprily be**

Chapter 2

Literature survey

Discussion and comparison of literature survey.

2.1 Introduction to overall topic (generate background for literature survey with reference to minimum 15 references)

2.2 Exhaustive literature survey of atleast 10 good journal or conference papers related to topic preferably starting from reference 1 to 15 given in the reference section serially. Literature survey should lead finally to identifying the gap in the research area and hence definition of problem statement.

Note:

1. **Please include the citations for the papers referred**
2. **Include it in the reference list at the same location as the number in the text.**

What is a literature review?

A literature review summarizes and synthesizes the existing scholarly research on a particular topic. Literature reviews are a form of academic writing commonly used in the sciences, social sciences, and humanities. However, unlike research papers, which establish new arguments and make original contributions**, literature reviews organize and present existing research.** As a student, you might produce a literature review as a standalone paper or as a portion of a larger research project.

Literature Review should be written in paragraph pattern in report and in comparative form in presentation.

Chapter 3

Methodology and Implementation

4.1 Block diagram

4.2 Hardware description

4.3 Software description, flowchart / algorithm

This chapter can comprise of actual implementation photos and their description.

Chapter 4

Results and Analysis

This shall include a thorough evaluation and investigation carried out. It should also bring out your contributions from the study. The discussion shall logically lead to inferences and conclusions as well as scope for possible further future work.

Note:

Include the IEEE or any other standards that you have adhered to test the validity of the results.

Link for IEEE standards

https://www.ieee.org/content/ieee-org/en/standards/index.html/

Chapter 5

Advantages, Limitations and Applications

Chapter 6

Conclusion and Future Scope

- A brief report of the work carried out, conclusions derived from logical analysis presented in the Results and Discussions chapter.
- Scope for future work should be stated lucidly in this chapter.

References

- Number all the references.
- Use a chronological bibliography.
- Each listed reference in the bibliography must be cited in the text of the report.
- For a book give the name(s) of author(s), title of book, edition, chapter number, and page numbers, publisher, location and year of publication.

- Example:
  [25] Jones, C.D., A.B. Smith, and E.F. Roberts, *Efficient Real-Time Fine-Grained Concurrency*, 2nd Ed., Ch. 3, pp. 145-7, Tata McGraw-Hill, New Delhi, 1994.

- For a journal/conference paper, give the name(s) of authors, title of paper, name of journal/ conference, volume and issue number (for journal), page numbers, and month and year of publication.
- Example:
  [23] Prasad, A.B., Kumar, C.D., Jones, E.F., and Frost, P.: "Cable Television Broadband Architectures", *IEEE Comm. Magazine*, vol. 39, pp. 134-141, June 1991.

Appendix A: Soft Code Flowcharts

Appendix B: Data Sheets

Appendix C: List of Components

Appendix D: List of Paper Presented and Published

- List of papers on the topic of the report published by the candidates.
- This may also be included in the contents.
- The candidates may also include reprints of his/her publications after the literature citation.